

Exploratory Data Analysis (EDA) Report

1. Introduction

Exploratory Data Analysis (EDA) is a crucial step in any machine learning project. This report presents an analysis of the cryptocurrency historical prices dataset used for volatility prediction. The objective of EDA is to understand the structure, distribution, trends, and relationships within the data before applying machine learning models.

2. Dataset Overview

The dataset contains historical daily records of multiple cryptocurrencies. Each record represents market information for a particular cryptocurrency on a specific date.

Dataset Characteristics:

- Number of records: Large-scale historical dataset
 - Data type: Time-series
 - Key features:
 - Open price
 - High price
 - Low price
 - Close price
 - Trading volume
 - Market capitalization
 - Cryptocurrency name
 - Date
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3. Data Cleaning Process

The following data cleaning steps were performed:

- Removed unnecessary index columns
- Converted the date column into datetime format
- Sorted data by cryptocurrency name and date
- Handled missing values using forward fill (ffill) to preserve time-series continuity
- Removed invalid or infinite values generated during feature engineering

These steps ensured consistency and reliability of the dataset.

4. Feature Engineering Summary

Additional features were created to enhance volatility prediction:

- Log returns
- Rolling volatility (7-day window)
- Price range (High – Low)
- Close–Open price difference
- Volume to market capitalization ratio
- Moving averages (7-day and 14-day)

5. Data Visualization and Insights

Price Trend Analysis

Line plots were used to visualize cryptocurrency price movements over time. The plots indicate strong fluctuations, confirming the volatile nature of cryptocurrency markets.

Volatility Distribution

Histogram plots of volatility show a right-skewed distribution, indicating frequent low volatility with occasional extreme spikes.

Correlation Analysis

Correlation heatmaps reveal strong relationships between price-based features such as open, high, low, and close prices.

6. Key Observations

- Cryptocurrency prices exhibit high volatility over time
- Trading volume varies significantly across assets
- Volatility tends to cluster during unstable market conditions

7. Conclusion

EDA helped in understanding market behavior, identifying trends, and selecting meaningful features for machine learning. The insights gained guided effective feature engineering and model selection.